

RM10-O-M02-BODY

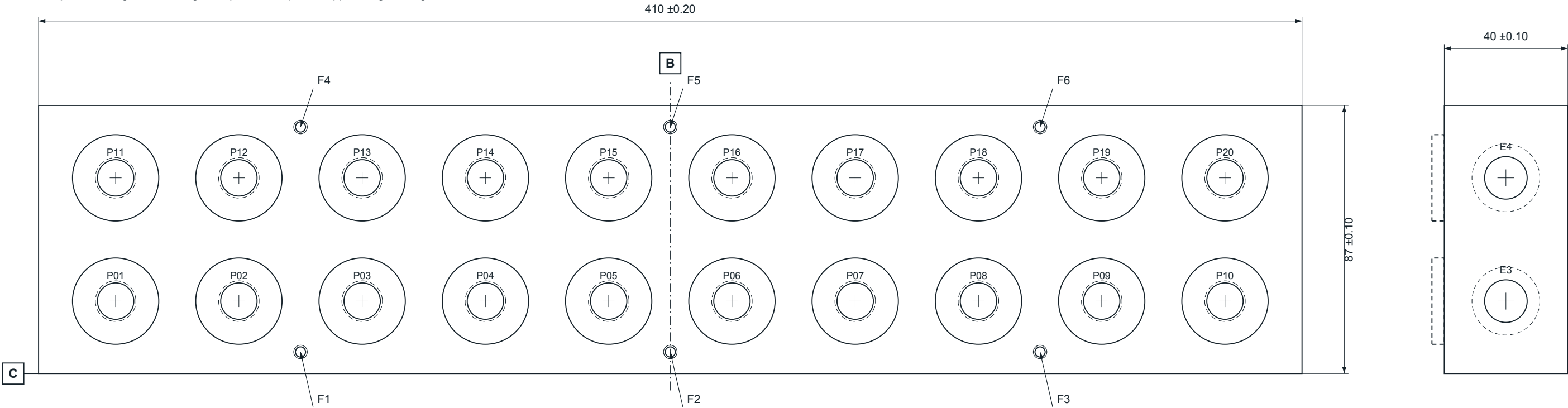
POM MANIFOLD — FRONT AND END PORT / FIXING LOCATIONS

LAYOUT O • DETAIL ISSUE O-M02

Single-part machining drawing

30 TAPPED FEATURES TOTAL: 20 front G 1/4 + 4 end G 1/4 + 6 M4 × 0.7. No thread inserts.

Two uninterrupted Ø11.8 galleries. Long-bore process requires supplier engineering confirmation before manufacture.



FRONT VIEW — looking +Y; countersinks / bosses face the viewer

RIGHT END  
looking -X

| FRONT PORT IDs | X ±0.10 | Z ±0.10     |
|----------------|---------|-------------|
| P01 / P11      | -180    | 23.5 / 63.5 |
| P02 / P12      | -140    | 23.5 / 63.5 |
| P03 / P13      | -100    | 23.5 / 63.5 |
| P04 / P14      | -60     | 23.5 / 63.5 |
| P05 / P15      | -20     | 23.5 / 63.5 |
| P06 / P16      | 20      | 23.5 / 63.5 |
| P07 / P17      | 60      | 23.5 / 63.5 |
| P08 / P18      | 100     | 23.5 / 63.5 |
| P09 / P19      | 140     | 23.5 / 63.5 |
| P10 / P20      | 180     | 23.5 / 63.5 |

| M4 IDs | X ±0.10 | Z ±0.10 |
|--------|---------|---------|
| F1     | -120    | 7       |
| F2     | 0       | 7       |
| F3     | 120     | 7       |
| F4     | -120    | 80      |
| F5     | 0       | 80      |
| F6     | 120     | 80      |

A = body front mounting plane Y0.  
B = width midplane X0; C = bottom Z0.  
Boss faces: Y-4.0 ±0.10 from A.  
Overall depth: 44 REF including bosses.

| END ID | FACE  | Y / Z ±0.10 |
|--------|-------|-------------|
| E1     | LEFT  | 20 / 23.5   |
| E2     | LEFT  | 20 / 63.5   |
| E3     | RIGHT | 20 / 23.5   |
| E4     | RIGHT | 20 / 63.5   |

Left face: X-205; right face: X+205.  
Both ends have the same Y/Z coordinates.  
Ø22 dashed rings show sealing lands;  
they are NOT recesses or grooves.

20 bosses: Ø28 ±0.10, root R1.0 ±0.10, outer lip C0.5 ±0.10 × 45° ±1° (see sheet 2).  
Front drill/thread axes normal to A; end port axes parallel to X. All threads right-hand.  
Surface requirements, bore controls, inspection and manufacturing notes: sheet 3. No product markings.

BLACK UNFILLED POM / C OR H

Units: mm • DIN ISO 2768-1 m unless stated

Dimensions at 20 °C • Do not scale drawing

Geometry: approved Revision O

Detail issue: O-M02 • 2026-09-15

Unmodelled helices: machine threads to this drawing

Scale: 0.73:1

Third-angle; views labelled

No product markings

SHEET 1 / 3

PDF + matching STEP

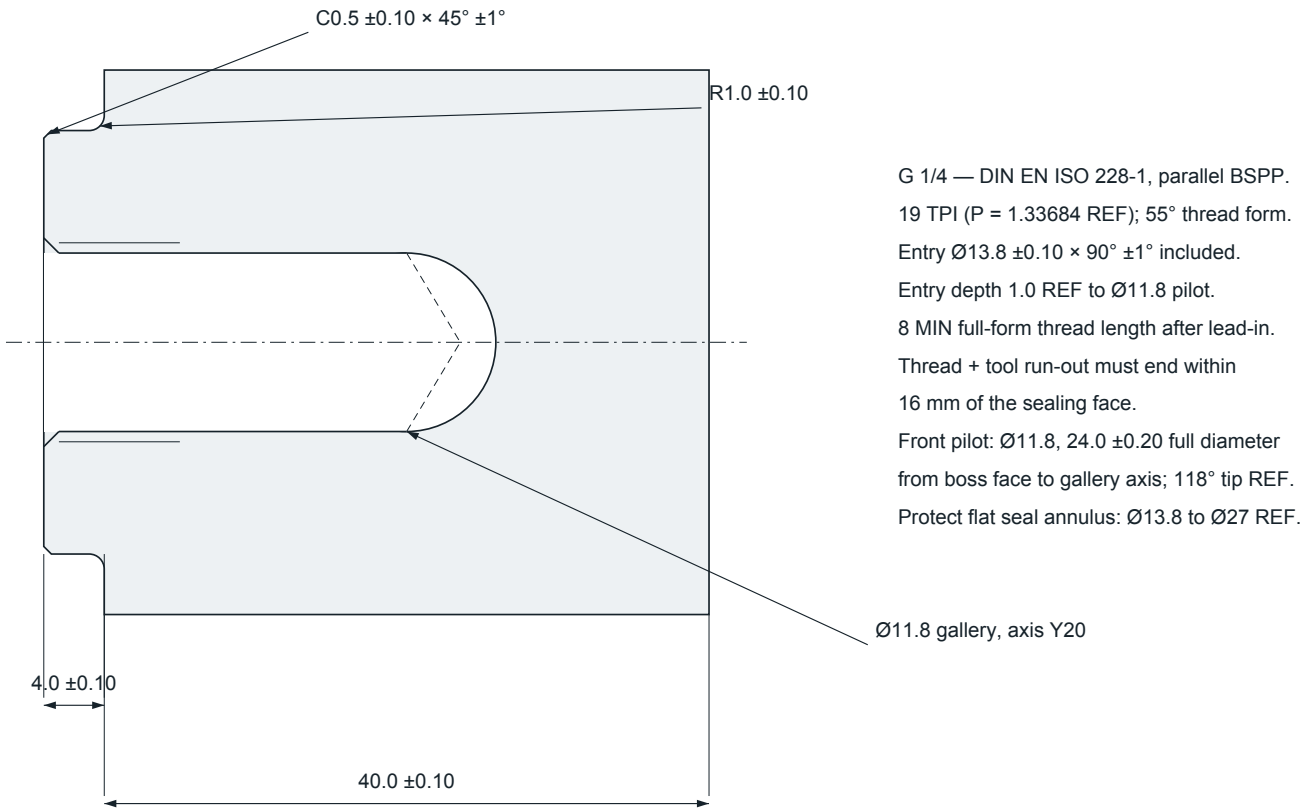
Quotation / supplier DFM

RM10-O-M02-BODY

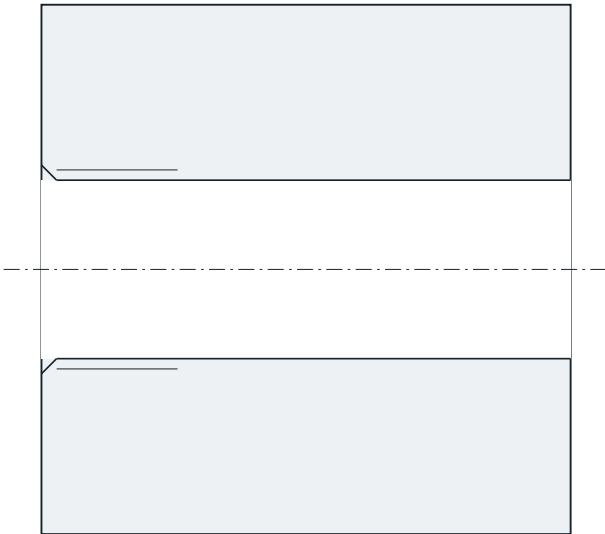
PORT, BOSS AND BLIND M4 DETAILS

Single-part machining drawing

DETAIL A — FRONT G 1/4 PORT / BOSS, 2:1 (20 places)

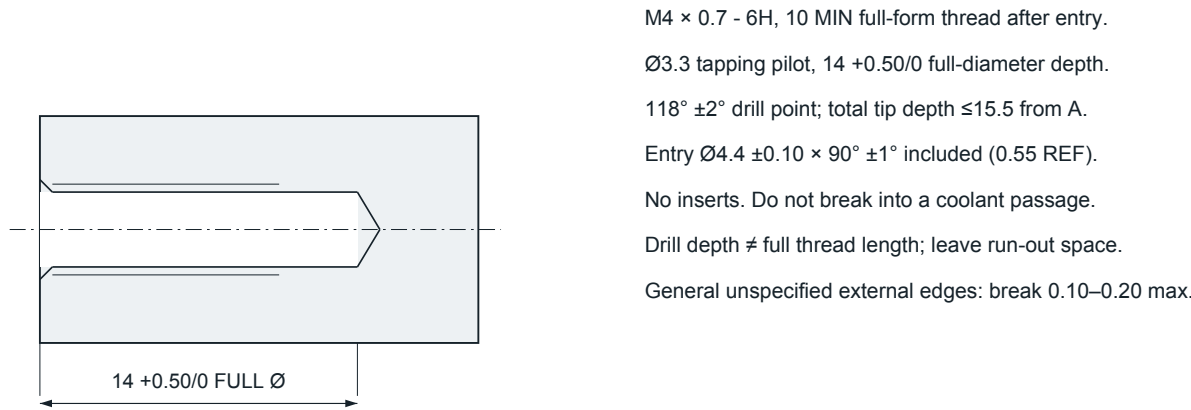


DETAIL B — END G 1/4, 2:1 (4 places)



Same G 1/4 thread and Ø13.8 × 90° entry.  
8 MIN full-form thread after lead-in.  
Thread/run-out ≤16 from each end face.  
Flat Ø22 MIN sealing land on end face.  
Unthreaded bore continues to opposite end.  
No end caps, inserts, glued plugs or splits.

DETAIL C — M4 × 0.7 - 6H (6 places), 3:1



BLACK UNFILLED POM / C OR H

Units: mm • DIN ISO 2768-1 m unless stated

Dimensions at 20 °C • Do not scale drawing

Geometry: approved Revision O

Detail issue: O-M02 • 2026-09-15

Unmodelled helices: machine threads to this drawing

Scale: AS SHOWN

Third-angle; views labelled

No product markings

SHEET 2 / 3

PDF + matching STEP

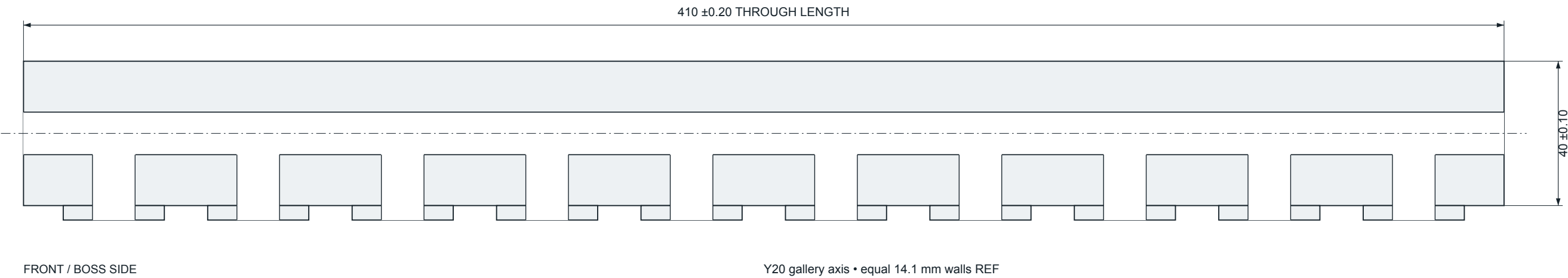
Quotation / supplier DFM

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LONG-BORE SECTION, TOLERANCES AND MACHINING NOTES

Single-part machining drawing

SECTION THROUGH ONE GALLERY AT Z23.5 — second gallery identical at Z63.5



BORES / THREADS / CLEANLINESS

- Two galleries Ø11.8 +0.08/0 through 410 mm, axes Y20; Z23.5 and Z63.5. Continuous parallel networks only.
- Opposed drilling is permissible. Nominal reach 207 mm per end gives 4 mm overlap; select tooling/process.
- Gallery centreline deviation ≤0.30 from nominal axis; meeting step ≤0.30. No blind webs or wall breakthrough.
- Unthreaded gallery / branch bores: Ra ≤6.3 µm.
- G 1/4: gauge to ISO 228-2. M4: gauge to ISO 1502.  
Thread quality governs the final tapped pilot diameter.  
G is PARALLEL; no NPT, R, Rp or tapered substitute.
- INTERNAL EDGE DEBURRING IS NOT REQUIRED.  
Clean / flush out loose machining chips only.  
Verify both galleries are continuous and open.
- Deliver dry and clean. No chemical treatment, impregnation or adhesive repairs without agreement. No product marks.

DATUMS / FUNCTIONAL SURFACES

- A: front body mounting plane; flatness 0.15.  
B: derived width midplane. C: bottom plane.  
Front boss/thread and M4 positions: X/Z ±0.10.  
Side mouth positions: Y/Z ±0.10 at each end face.
- Boss OD Ø28 ±0.10; height 4.0 ±0.10; root R1 ±0.10.  
Lip C0.5 ±0.10 ×45° ±1°. Do not round the sealing face.
- Each front sealing annulus and each Ø22 end land:  
Ra ≤1.6 µm, local flatness 0.05; no radial scratches.  
Seal plane perpendicular to its thread axis within 0.05 across Ø22. Do not recess the sealing lands.
- Other machined exterior surfaces: Ra ≤3.2 µm.
- Dimensions apply after material stabilisation at 20 °C.  
Declare resin grade and stock form before manufacture.  
POM-C or POM-H accepted; supply grade datasheet.
- 2 mm rack-envelope gap is not a tolerance allocation; this drawing does not qualify the assembled rack fit.

SUPPLIER DFM HOLD POINT: confirm long-bore process, alignment inspection and material grade before cutting.

STEP contains nominal tapping pilots, not thread helices. General external edge deburring is not explicitly modelled.  
Boss lip chamfers, port-entry cones and M4 pilot drill points are modelled in the matching O-M02 STEP.  
Operator approval applies to layout O. This issue uses 4 mm bosses and edge / drill details; no pressure or structural rating is claimed.

BLACK UNFILLED POM / C OR H

Units: mm • DIN ISO 2768-1 m unless stated

Dimensions at 20 °C • Do not scale drawing

Geometry: approved Revision O

Detail issue: O-M02 • 2026-09-15

Unmodelled helices: machine threads to this drawing

Scale: SECTION 0.80:1

Third-angle; views labelled

No product markings

SHEET 3 / 3

PDF + matching STEP

Quotation / supplier DFM